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AI/NLP Financial Risk Engine

From headlines and posts to explainable risk signals, a sentiment-driven index and an event-driven portfolio stress test

Katariya Hemanth Kumar

Indian Institute of Technology Patna

Risk news never reaches the portfolio

The problem

- Analysts scan thousands of headlines and posts a day; coverage is slow and inconsistent
- A sentiment number alone does not say what happened or how much it matters
- A score nobody can explain will not pass model validation

One engine, three fields per signal

1

Sentiment score

-1 to 1, from FinBERT

2

Event classification

11 event types, from geopolitical to credit event

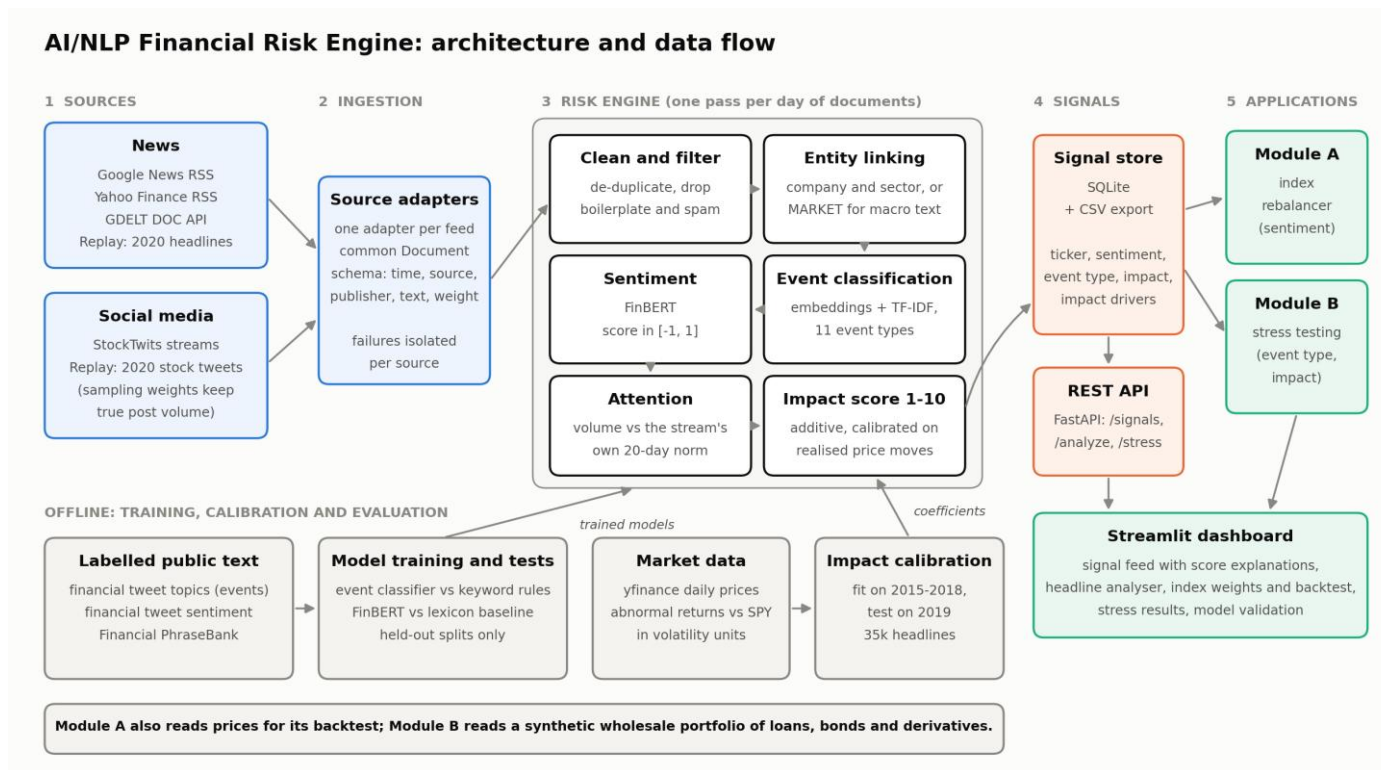
3

Impact score

1 to 10, fitted to how far prices really moved

Both downstream modules run on the same signal feed: Module A (index rebalancer) and Module B (portfolio stress test).

From raw text to portfolio impact



Four choices that make signals trustworthy

1 Entity linking that knows roles

A bank quoted as the broker ("rating reaffirmed at JPMorgan") is not the subject. Social posts link by cashtag only.

2 Impact fitted to the market

An additive model trained on abnormal price moves after 35,000 headlines. Every score shows its drivers.

3 Attention measured honestly

Volume against each stream's own norm; share of voice for social, where the collector sets the volume.

4 Fully local and reproducible

Open models on CPU, no paid API. Score cache, replay and live modes, REST API, 45 automated tests.

Each model beats its naive baseline

71.6%

Sentiment accuracy
lexicon baseline 49.6%

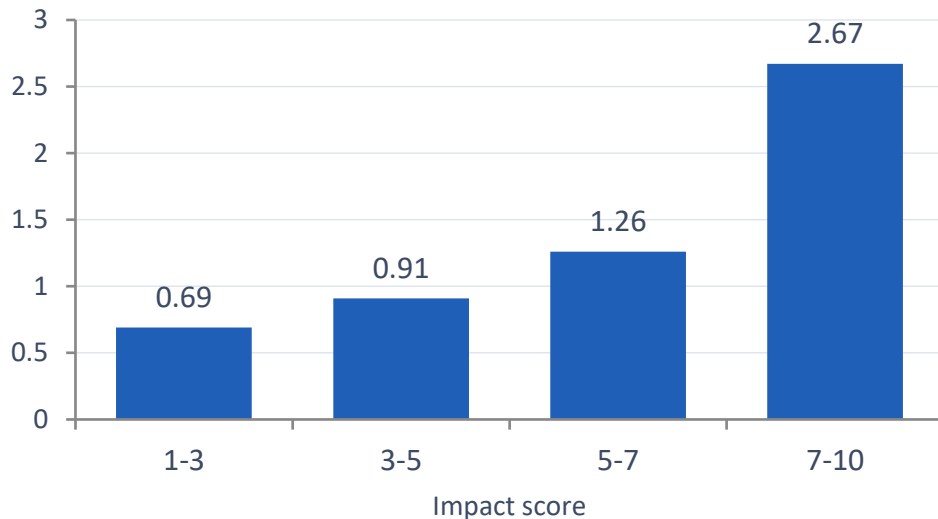
85.9%

Event classification
keyword rules 54.7%

38%

Big move when impact > 7
any stock-day 8%

Price move that followed, by impact score band



Impact tested on 5,969 headlines from 2019, fitted on 2015-2018. Move is the abnormal return over two days, in units of the stock's own volatility.

Signals priced two real stress days

Module B: stress testing

-\$162m

10 Mar 2020, after the oil-price crash

-\$135m

2 Apr 2020, record US jobless claims

- Triggered automatically on a \$7.4bn synthetic book of loans, bonds and derivatives
- Both days had the VIX near 50; credit spreads drove the loss

Module A: index rebalancer

+10.8%

sentiment-tilted index, Jan-Jul 2020

+11.2%

equal-weight index, same 18 stocks

- No outperformance, slightly lower volatility and drawdown
- Sentiment tracks the same day's return but not the next tradable one: it describes moves, it does not forecast them

What is not yet proven, and what is next

Limitations

- Market-wide impact is not validated: the public headlines are stock commentary, not a macro newswire
- Coverage volume drives most of the impact score; event type adds little
- One six-month window that includes the COVID crash
- Stress model is a simplified sensitivity revaluation on a synthetic book

Next steps

- 1 Time-stamped newswire or GDELT events for the market scope
- 2 Sentiment per company inside one headline
- 3 A continuous live poller, so attention is measured on live data
- 4 Rating migration and correlated shocks in the stress engine